Mustafa Ahmed

Applied AI Engineer | Backend Engineer | LLM Systems
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Summary

Applied AI Engineer with 3.5+ years of backend engineering experience, building AI agents for regulated work: governed tool access, human review gates, evaluation and audit trails. Built Warden, an identity and access gateway for AI agents, and Tickmark, an AI audit agent with an Excel add-in that vouches invoice samples against invoice PDFs. Currently building Standwise, a multi-tenant SaaS work platform with Claude-powered AI standups. Builds in Python, TypeScript and Go.

Featured Projects

Warden: Identity and Access Gateway for AI Agents | *C#, .NET 10, ASP.NET Core, PostgreSQL, Microsoft Entra ID, MCP, Azure Container Apps*

- Built an MCP gateway that sits between any AI agent (Claude, VS Code or custom) and a firm's tool servers, authorizing every tool call against the firm's rules with no changes to the agent.
- Designed a fail-closed decision engine that runs 7 checks on every call (firm, membership, ethical wall, role, agent ceiling and tool scope); agents act for a signed-in Microsoft Entra user through 15-minute, engagement-scoped RS256 passes and can never exceed that user's permissions.
- Blocked prompt injection by hiding forbidden tools from the agent entirely, and locked tool servers behind 1-minute, single-tool tokens with no public address in Azure.
- Built an append-only audit log enforced by PostgreSQL triggers, plus SCIM-based offboarding that revokes all of a departing user's agent passes immediately.
- Shipped a working MVP with 74 xUnit tests named as security promises, each verified by deliberately breaking the check it covers; proved real Entra sign-in end to end with Claude Code.

Tickmark: AI Audit Agent for Invoice Vouching (demo agent behind Warden) | *Python 3.12, FastAPI, PostgreSQL, Azure Document Intelligence, Office.js*

- Built an AI agent that checks Excel invoice samples against invoice PDFs, marking each row matched, difference, needs review or no evidence with a written reason; designed so it never marks a row matched when it is unsure.
- Kept AI out of every decision ("AI reads, code decides"): the LLM only suggests what each spreadsheet column means, while code makes every call using Decimal money with ±0.01 tolerance, VAT rate, audit period, duplicate and line-total checks.
- Built a hybrid extraction pipeline: pdfplumber reads PDFs locally and only unclear ones go to Azure Document Intelligence; after measuring that extractor confidence scores did not separate right from wrong values, a value counts as confirmed only when the arithmetic checks out or it is printed on the page.
- Built an append-only, multi-tenant audit store in PostgreSQL: triggers block UPDATE, DELETE and TRUNCATE, row-level security isolates each firm and client engagement (deny by default), and every file is stored by SHA-256 hash so any run can be re-checked.
- Shipped an Excel add-in (Office.js) that writes colour-coded verdict columns and opens the matching invoice on row click, plus a web app with live progress.
- Secured the command-line agent through Warden: it swaps the user's token for a 15-minute pass limited to one client and two actions, and its tools accept only 60-second RS256 tokens verified against Warden's JWKS.
- Scored 30/30 on real ZUGFeRD e-invoices and 34/40 on varied invoice originals, backed by 171 pytest tests; CI fails the build on any false match or wrong evidence link.

Experience

AI Engineer (Solutions) – *ByteForge* Dec 2025 – Present

- Architected and built Standwise, a multi-tenant, chat-first SaaS work platform (HR and project management) as a modular TypeScript monolith on NestJS, PostgreSQL, Redis and AWS ECS Fargate, running with ByteForge as its first tenant.
- Built an AI standup engine on the Claude API that routes tasks across Haiku, Sonnet and Opus models, turns free-text updates into structured data, flags implied blockers, and posts summaries and lead alerts to Slack.
- Designed a "code decides, LLM writes" architecture so the model never owns facts: sprint detection, risk rules, metrics and permissions run in tested code, and the LLM only generates the language.
- Shipped LLM guardrails: zod-validated structured outputs with retry, versioned prompts, redaction of sensitive HR data, human approval before any AI write to Jira or calendars, and audit logs of every agent action.
- Added CI evaluations with number-grounding hallucination checks that block merges on regression.

- Implemented three-layer tenant isolation (query scoping, PostgreSQL row-level security, CI isolation test suite) with scoped RBAC and entitlement-gated features.

Founder & Software Engineer – *Khel* Jan 2026 – Present

- Designed and built Khel end to end, a pre-launch turf-booking platform for Pakistan: product, Go backend, React Native mobile app, payments and matchmaking.
- Built the reservation system, handling booking conflicts, payment deadlines and partial payments.
- Built a peer-to-peer split-payment flow so players can share a booking's cost without a third-party payment gateway.
- Built an ELO-based player rating system for skill-based matchmaking, plus badges and streaks for player engagement.

Software Engineer – *Infoetech* Nov 2023 – Dec 2025

- Built RESTful APIs and microservices (Node.js, Express, Socket.IO) powering search, availability and checkout for a tour-booking platform.
- Improved API performance through database index tuning and Redis caching.
- Ran zero-downtime AWS deployments with Docker and Kubernetes.

Software Engineer – *Ethisol* Mar 2023 – Nov 2023

- Built backend services in Node.js and MongoDB with Stripe and PayPal payment integration.
- Implemented JWT authentication and role-based access control (RBAC) for enterprise client applications.

Skills

AI & LLMs: Large Language Models (LLMs), AI Agents, Multi-Agent Systems, Model Context Protocol (MCP), Anthropic Claude API, LangGraph, LangChain, RAG, pgvector, Chroma, Prompt Engineering, Structured Outputs, Agent Evaluation, Document Extraction (pdfplumber), Prompt-Injection Testing, Guardrails, Human-in-the-Loop Approval, Risk-Tiered Permissions
Observability: OpenTelemetry, Prometheus, Grafana, Tempo, Langfuse
Languages: Python, TypeScript, JavaScript, Go, C#, SQL
Backend: FastAPI, NestJS, .NET, ASP.NET Core, EF Core, Node.js, Express, Socket.IO, REST APIs, Microservices, JWT, RBAC, Stripe, PayPal
Identity & Security: Microsoft Entra ID, OAuth 2.0, PKCE, JWT, SCIM, RBAC, Multi-Tenant Isolation, Audit Logging
Databases: PostgreSQL (Row-Level Security), MongoDB, Redis, Prisma, Alembic, sqlx
Cloud & DevOps: AWS (ECS Fargate, EKS, S3, SQS, RDS, SES), Azure (Container Apps, PostgreSQL Flexible Server, Key Vault, Bicep, Document Intelligence), Docker, Kubernetes, Terraform, Linux, GitHub Actions, CI/CD, pytest, Git
Frontend & Mobile: Office.js (Excel Add-ins), React Native

Other Projects

LLM Gateway | *Python, FastAPI, Prometheus, OpenTelemetry, Tempo, Grafana*

- Built a FastAPI proxy for the Claude API with per-tenant authentication, rate limiting and token quotas.
- Instrumented it end to end: OpenTelemetry traces exported to Tempo, Prometheus metrics, and Grafana dashboards for request latency, token usage and errors per tenant.

Compliance Copilot | *Python, FastAPI, PostgreSQL, Chroma, Langfuse*

- Built a multi-tenant AI agent that checks each organization's GitHub and AWS setup against SOC 2, GDPR, HIPAA, PCI DSS and NIST CSF controls, using RAG over policy documents, and cites the evidence behind every finding.
- Drafts a fix for each gap and never applies a change without human approval; every agent run is traced in Langfuse.

Source code for all projects: github.com/musishere

Education

Bachelor of Computer Science – *University of Lahore*